

ADS-B Out MoC review

AC 500-029 Issue 2 Transmittal Performance requirements

AEA conference 12 September 2025



Transport
Canada

Transports
Canada

Canada

Presentation Scope

- Focus of discussion today
 - Antenna installation requirements AWM 551.103(d)(4)
 - Capacity/performance to transmit to top and bottom
 - NAV CANADA test flight quantitative evaluation (Probability of Update PU)
 - The most “contentious” MoC from feedback received
- Other requirements from AWM 551.103
 - Check AC 500-029 Issue 02
 - Check former familiarization material – (Feb 2024)



Antenna installation requirement

AWM 551.103(d)(4): antennas [...]shall be installed so that:

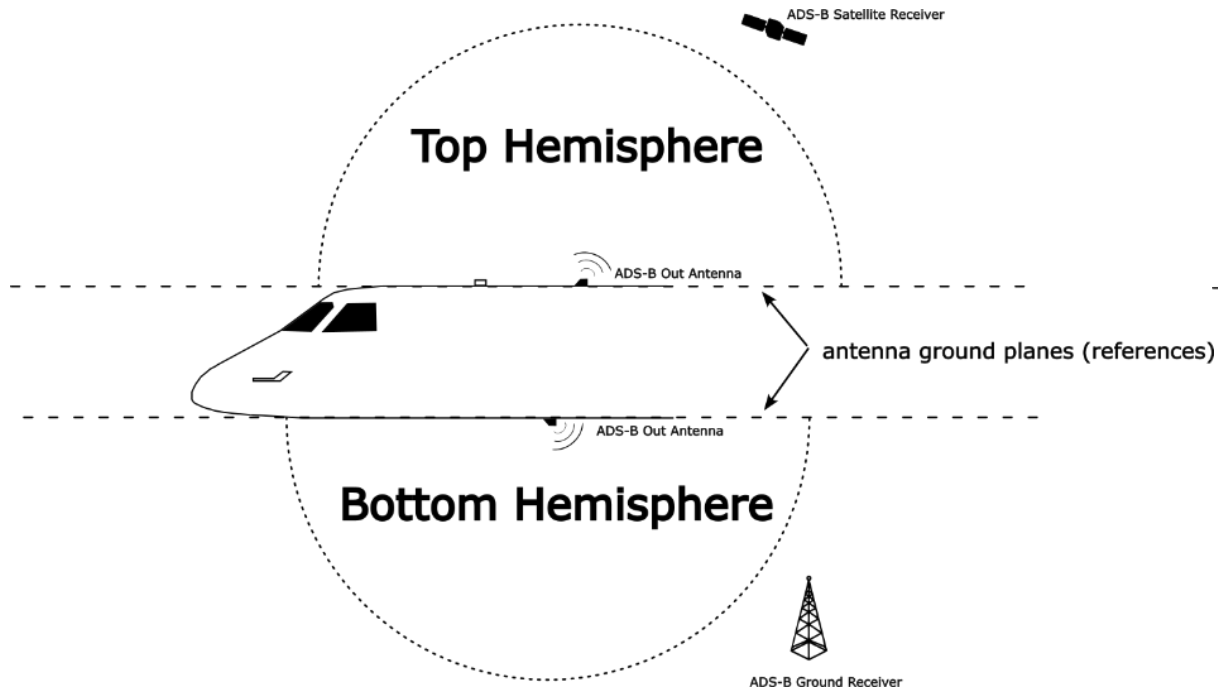
(Option 1)

i. For Class A1, A2, A3 or B1 ... they comply with ...section 3.3 of RTCA DO-260B ...and **they transmit alternately towards the bottom and the top** ...per the requirements ... in section 2.2.13.6 (“transmitting diversity”); **or**

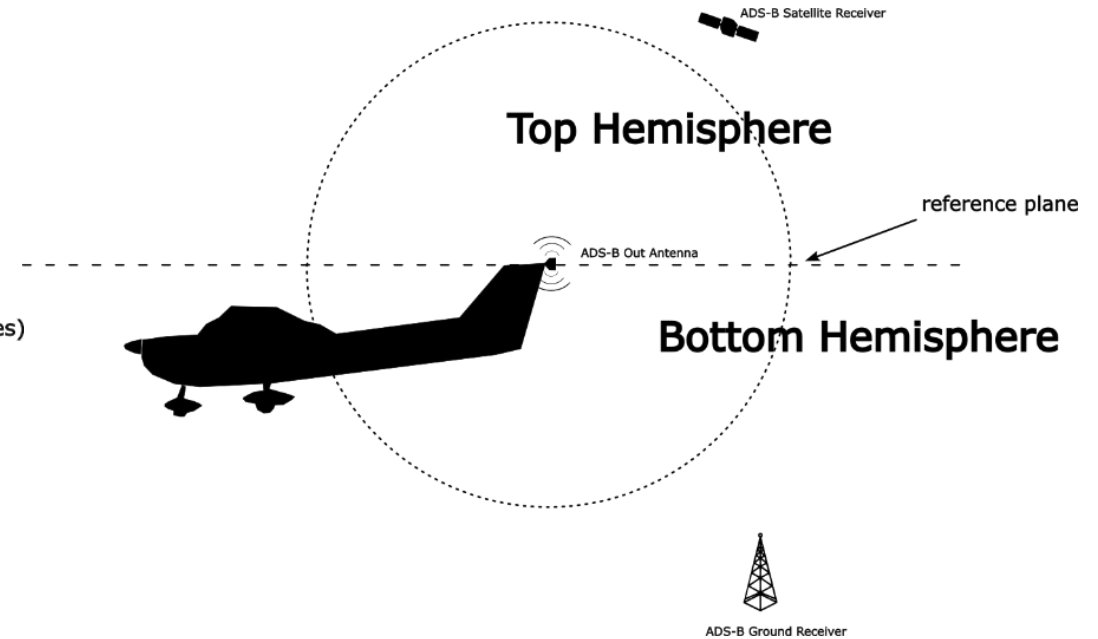
(Option 2)

ii. For Class A1S or B1S ...they comply with ...section 3.3 of RTCA DO-260B ...**and it can be demonstrated that the equipment transmits satisfactorily to ground-based as well as space-based receiver**...

Key verification: Transmittal towards top and bottom



Diversity



Non-diversity

Performance test flight

- **End goal of requirements: support separation standards per ED-129**
 - Methodology and metrics defined in ED-129B
 - Measures minimum performance as defined in ED-129B – Probability of Update
- **Direct performance measurement of end goal**
 - Higher level requirement than AWM 551.103(d)(4)(i)
 - Measures receiving rate at satellite (vs transmittal rate at antenna)
 - Performance report compiled and supplied by NAV CANADA
 - “A” test flight is necessary regardless (other installation factors to consider vs ground test): complete system check in operational environment

Quantitative end level test flight deemed easier and more encompassing

Performance measurement DAPR vs PAPR (1)

DAPR = Detailed ADS-B Performance Report [NAV CANADA Public ADS-B Performance Report](#)

- Reports Probability of Update (PU) + other accuracy and integrity metrics
- Data from Aireon (Satellites): all messages received manually collected and analyzed upon request for calculation of PU.
- No automatic generation: must be requested prior to flight
- Designed to be used for certification projects only

PAPR = Public ADS-B Performance Report

- Reports Probability of Update (PU) + other accuracy and integrity metrics
- Aireon Data automatically generated, throttled every 4s (not all messages received) → less accurate in terms of PU (see next slide)
- Can be easily obtained after flight from NAV CANADA's site
- Suitable for self-assessment, not certification, due to lower accuracy

Performance measurement

DAPR vs PAPR (2)

- DAPR: all messages received downloaded and used in calculation of PU
- PAPR: Received messages throttled at 4 s: some received messages in the relevant interval (5s, 8s) may be missed in calculation of PU.

Consequences:

- For the same flight, performance reported by PAPR will always be lower than, or equal to actual performance
- The lower the performance, the higher the gap
- The higher the PU interval (e.g. 8s, 30s, vs 5s) the lower the impact/gap
- PAPR may show substantially lower PU performance vs DAPR in lower performance scenarios
- PAPR performance numbers only relevant for high PUs (> 90% TBC by NC)

PAPR is a good tool to estimate compliance only

Performance criteria

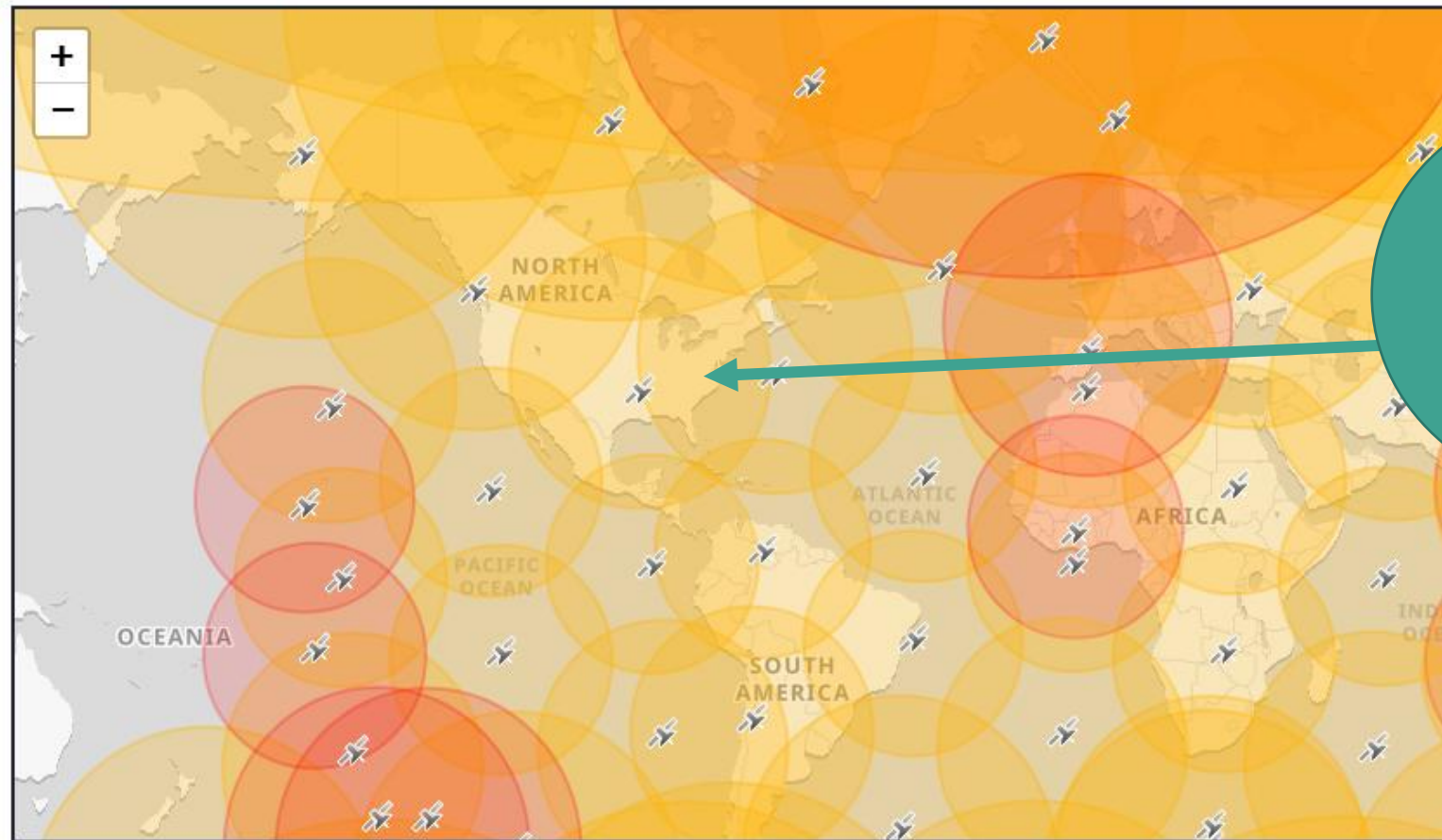
The PU interval selection issue

- **End goal: support separation standards per ED-129()**
 - Methodology and metrics defined in ED-129B
 - AC 500-029 Issue 02, sec. 6.4(11): Minimum performance: **Probability of Update at 5 seconds (PU_{5s}) $\geq 96.5\%$**
 - *Supports separation standards of 3 nm for all airspace (including terminal area) based on ED-129B*
 - *DAPR now provides PU with interval of 8s: updated to support separation of 5 nm (enroute, applicable to Class A and B) per ED-129C → mismatch with criteria in AC 500-029. To be updated.*
 - *Amplifies discrepancy with PAPR report (PAPR still reports PU at 5s interval). $PU_{5s} < PU_{8s}$*
- **Next steps:**
 - Update AC 500-029 Probability of Update at 8 seconds vs 5 seconds (PU_{8s}) $\geq 96.5\%$
 - NAV CANADA: request to update PAPR to report PU_{8s}

For Class A and B airspace, PU interval to be adjusted to 8s in AC 500-29

Performance criteria

The congestion issue (1)



Sat swath is
an ellipse
roughly
3000km wide

<https://www.groundcontrol.com/en/knowledge/calculators-and-maps/iridium-coverage-map-real-time-tracking/>

Performance criteria

The congestion issue (2)

wide swath = Closest satellite (overhead) receives messages simultaneously from many aircraft when test aircraft located in a high traffic density zone.

→ Inconsistent message reception for some aircraft,
degraded PU performance

!!!!Notes:

- 1) NOT an issue from a surveillance point of view. These areas are covered by other surveillance methods (e.g. radar)
- 2) Only an issue for this flight testing as it can affect the pass status

Performance criteria

The congestion issue (3)

Solutions for a PU performance-based test:

1) Perform test in low traffic density conditions (see next slide)

2) compare test aircraft message PU performance *at satellite*
versus peer aircraft in the vicinity → DAPR presents the
comparative analysis

Issues:

- Subject to other aircraft performance
- Comparative statistical approach only. No definitive performance measurement
- NAV CANADA solutions? Alternate approaches?

Performance criteria

The congestion issue (4)

Perform test in low traffic density conditions

2 main factors:

1) Time

Avoid time of high traffic (night) → not always practical

2) Location

Avoid testing in “red zone”.

- Red zone contours illustrated in PAPR User Guide



PAPR User Guide

Performance criteria

The congestion issue (5) – Red zone

For illustration
Possible to publish zone depending on time of
day? Dynamically? More precise
specification?



Request NAV CANADA to define red zone?

Performance criteria

Alternative MoC

Other means of compliance may be acceptable

- Keep in mind requirement: transmit to top and bottom
- Exploring other flight tests based on signal received by other aircraft (vs received at satellites)

Thank you